Modelling and Forecasting Realized Variance

Financial Econometrics – Homework 4
MSc in Data Sciences and Business Analytics

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1 Introduction

We study daily *Realized Variance* (RV) for the 30 constituents of the Dow Jones Industrial Average using high-frequency intraday data spanning January 2003 to March 2022 (≈4 800 trading days per stock). RV is constructed as the sum of squared 5-minute log-returns:

$$RV_t = \sum_{s=1}^{n} r_s^2, \quad r_s = \log\left(\frac{P_s}{P_{s-1}}\right) \tag{1}$$

Five-minute sampling balances microstructure noise against information content. We compare econometric and machine learning (ML) forecasting models and assess cross-stock predictability via Granger causality.

2 Data and Descriptive Analysis

Table 1 reports descriptive statistics for $\ln RV$ for four representative stocks. Raw RV is right-skewed and leptokurtic; the log-transform substantially reduces both. Figure 1 shows the $\ln RV$ series for AAPL — volatility spikes during the GFC (2008) and COVID-19 (2020) are clearly visible. Sample ACFs of $\ln RV$ decay very slowly (Figure 2), confirming long memory. Augmented Dickey–Fuller tests reject the unit-root null for **all 30 stocks** at the 5% level, confirming stationarity of $\ln RV$.

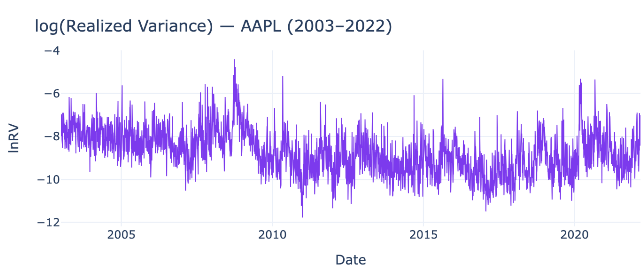

Figure 1: $\ln RV$ for AAPL (2003–2022).

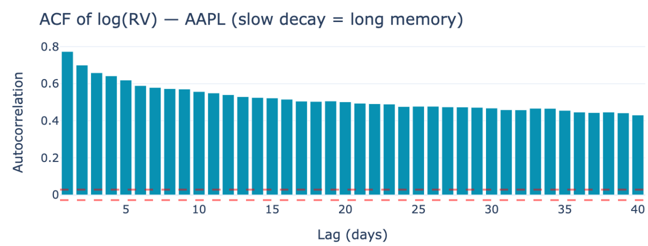

Figure 2: Sample ACF of $\ln RV$ for AAPL. Slow decay confirms long memory.

Table 1: Descriptive statistics of $\ln RV$ (selected stocks)

Ticker	Mean	Variance	Skewness	Kurtosis
AAPL	−8.74	0.96	0.31	0.21
JPM	−8.88	1.09	1.10	1.73
MSFT	−9.08	0.69	0.79	1.40
GS	−8.76	0.85	1.23	2.78

3 Models

HAR (Corsi, 2009).

$$y_t = \beta_0 + \beta_d y_{t-1} + \beta_w \bar{y}_{t-1}^{(5)} + \beta_m \bar{y}_{t-1}^{(22)} + \varepsilon_t \tag{2}$$

where $\bar{y}_{t-1}^{(n)} = n^{-1}\sum_{i=1}^{n} y_{t-i}$. This is a restricted AR(22) estimated by OLS, capturing heterogeneous trader horizons. HAR coefficients are positive and economically meaningful across all 30 stocks, with $\beta_d > \beta_w > \beta_m$ on average and $R^2 \approx 0.65$.

Benchmarks. AR(1) and a random walk ($\hat{y}_{t+1} = y_t$).

Machine Learning. Four ML models use the same three HAR predictors for a fair comparison: **LASSO** (L1-regularised), **Ridge** (L2-regularised), **XGBoost** (gradient-boosted trees), and **LightGBM** (histogram-based boosting). All ML models are retrained annually (every 252 trading days), following the approach of [1].

4 Out-of-Sample Forecasting

Models are evaluated on the last 50% of observations (horizon $h = 1$, annual retraining). Table 2 reports average RMSE and MFE across all 30 stocks.

Table 2: Average RMSE and MFE across 30 DJIA stocks (OOS, $h = 1$)

Model	Avg. RMSE	Avg. MFE
Ridge	**0.5107**	0.0147
HAR	0.5108	0.0137
LASSO	0.5108	0.0140
XGBoost	0.5207	0.0288
LightGBM	0.5235	0.0270
AR(1)	0.5492	0.0680
Rand. Walk	0.5793	−0.0004

Key findings: (i) HAR, LASSO, and Ridge are statistically indistinguishable at the top, confirming the difficulty of beating a well-specified linear benchmark [1]. (ii) Tree-based ML models (XGBoost, LightGBM) rank fourth and fifth – competitive but offering no consistent improvement over HAR. (iii) The Random Walk is always worst, confirming significant predictability in $\ln RV$. (iv) All models are positively biased (MFE > 0), except the Random Walk which is nearly unbiased. Figure 3 summarises these results visually.

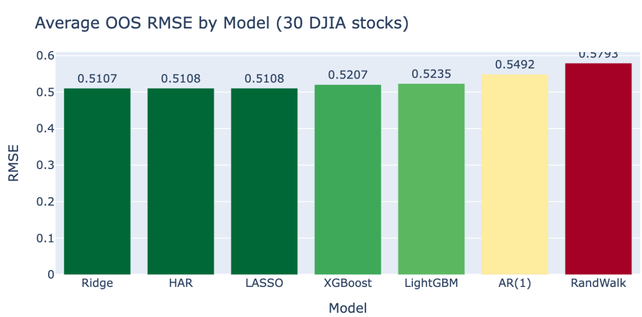

Figure 3: Average OOS RMSE across 30 DJIA stocks. HAR, LASSO, and Ridge are jointly best; the Random Walk is always worst.

5 Granger Causality

We test all $30 \times 29 = 870$ directed pairs for Granger causality using the minimum F-test p-value across lags 1–10. **869 out of 870 pairs** (99.9%) are significant at the 5% level (Figure 4), indicating pervasive cross-stock volatility spillovers across the DJIA.

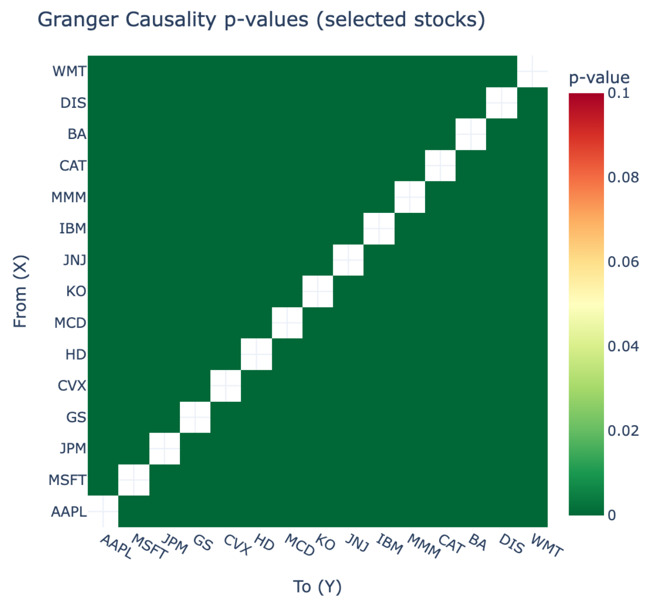

Figure 4: Granger causality p-values for 15 selected DJIA stocks. Red = highly significant ($p \approx 0$); nearly all pairs are significant, reflecting strong index co-movement.

For the top-10 pairs by significance we run bivariate VAR forecasts (parameters estimated once on the training set). The VAR improves on the univariate HAR in **4 out of 10 cases**, with an average RMSE reduction of 0.004 for winning pairs. While Granger causality is statistically ubiquitous, its *incremental* forecasting value is modest – consistent with [1]'s finding that ML and richer models offer limited gains over parsimonious benchmarks.

6 Conclusion

HAR, LASSO, and Ridge are jointly the best forecasting models for $\ln RV$ across the DJIA, with near-identical RMSE of 0.51. Tree-based ML methods are competitive but offer no consistent advantage. Cross-stock Granger causality is statistically pervasive (869/870 pairs significant) yet adds limited out-of-sample forecasting value, with VAR improvements only modest (RMSE $\Delta \approx 0.004$). These results reinforce the conclusion of [1]: parsimonious linear econometric models remain the reliable benchmark for realized variance forecasting.

References

[1] R. Kilic (2025). *Linear and nonlinear econometric models against machine learning models: realized volatility prediction.* Finance and Economics Discussion Series, 2025-061.